

Binary Numbers — Create Level

Complete Answers

1. Creating Binary Numbers from Decimal Numbers

Decimal	Division Steps	Binary
17_{10}	$17 \div 2 = 8R1$, $8 \div 2 = 4R0$, $4 \div 2 = 2R0$, $2 \div 2 = 1R0$, $1 \div 2 = 0R1$	10001_2
24_{10}	$24 \div 2 = 12R0$, $12 \div 2 = 6R0$, $6 \div 2 = 3R0$, $3 \div 2 = 1R1$, $1 \div 2 = 0R1$	11000_2
35_{10}	$35 \div 2 = 17R1$, $17 \div 2 = 8R1$, $8 \div 2 = 4R0$, $4 \div 2 = 2R0$, $2 \div 2 = 1R0$, $1 \div 2 = 0R1$	100011_2
40_{10}	$40 \div 2 = 20R0$, $20 \div 2 = 10R0$, $10 \div 2 = 5R0$, $5 \div 2 = 2R1$, $2 \div 2 = 1R0$, $1 \div 2 = 0R1$	101000_2
50_{10}	$50 \div 2 = 25R0$, $25 \div 2 = 12R1$, $12 \div 2 = 6R0$, $6 \div 2 = 3R0$, $3 \div 2 = 1R1$, $1 \div 2 = 0R1$	110010_2

2. Creating Decimal Numbers from Binary Numbers

Binary	Place Values	Multiplication	Decimal
10111_2	16,8,4,2,1	$1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 1 \times 1$	23_{10}
11011_2	16,8,4,2,1	$1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$	27_{10}
10011_2	16,8,4,2,1	$1 \times 16 + 0 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$	19_{10}
11101_2	16,8,4,2,1	$1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$	29_{10}
101001_2	32,16,8,4,2,1	$1 \times 32 + 0 \times 16 + 1 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1$	41_{10}

3. Creating Binary Addition Problems

Simple Addition Answers

Problem	Column Addition	Answer
$100_2 + 10_2$	$1+0=1, 0+1=1, 1+0=1$	110_2
$110_2 + 101_2$	$0+1=1, 1+0=1, 1+1=2 \rightarrow 0 \text{ carry } 1$	1011_2
$1010_2 + 100_2$	$0+0=0, 1+0=1, 0+1=1, 1+0=1$	1110_2
$111_2 + 100_2$	$1+0=1, 1+0=1, 1+1=2 \rightarrow 0 \text{ carry } 1$	1011_2

Addition with Carrying Answers

Problem	Carries	Answer
$111_2 + 1_2$	1, 1, 1	1000_2
$1011_2 + 11_2$	1, 1, 0	1110_2
$1101_2 + 111_2$	1, 1, 1, 0	10100_2
$1111_2 + 111_2$	1, 1, 1, 1	10110_2

Multiple Addition Answers

Problem	Decimal Check	Answer
$10_2 + 11_2 + 1_2$	$2+3+1=6 = 110_2$	110_2
$111_2 + 11_2 + 1_2$	$7+3+1=11 = 1011_2$	1011_2
$101_2 + 11_2 + 10_2$	$5+3+2=10 = 1010_2$	1010_2
$110_2 + 101_2 + 11_2$	$6+5+3=14 = 1110_2$	1110_2

4. Creating Binary Subtraction Problems

Simple Subtraction Answers

Problem	Column Subtraction	Answer
$101_2 - 1_2$	$1-1=0$, $0-0=0$, $1-0=1$	100_2
$110_2 - 10_2$	$0-0=0$, $1-1=0$, $1-0=1$	100_2
$1000_2 - 10_2$	$0-0=0$, $0-1 \rightarrow$ borrow, etc.	110_2
$1010_2 - 100_2$	$0-0=0$, $1-0=1$, $0-1 \rightarrow$ borrow, $1-0=1$	110_2

Subtraction with Borrowing Answers

Problem	Borrow From	Answer
$1000_2 - 1_2$	Right column	111_2
$1100_2 - 1_2$	Right column	1011_2
$10000_2 - 1_2$	Right column	1111_2
$10100_2 - 11_2$	Right/middle columns	10001_2

5. Creating Conversion Tables

Binary	Step-by-Step	Decimal
101_2	$1 \times 4 + 0 \times 2 + 1 \times 1 = 4 + 0 + 1$	5_{10}
1101_2	$1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 8 + 4 + 0 + 1$	13_{10}
10110_2	$1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1 = 16 + 0 + 4 + 2 + 0$	22_{10}
11101_2	$1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 16 + 8 + 4 + 0 + 1$	29_{10}

6. Creating Worksheets

Worksheet 1: Decimal to Binary — Answer Key

Question	Answer
$8_{10} = \text{_____}_2$	1000_2
$16_{10} = \text{_____}_2$	10000_2
$29_{10} = \text{_____}_2$	11101_2
$34_{10} = \text{_____}_2$	100010_2
$47_{10} = \text{_____}_2$	101111_2

Worksheet 2: Binary to Decimal — Answer Key

Question	Answer
$111_2 = \text{_____}_{10}$	7_{10}
$1001_2 = \text{_____}_{10}$	9_{10}
$1111_2 = \text{_____}_{10}$	15_{10}
$10101_2 = \text{_____}_{10}$	21_{10}
$11011_2 = \text{_____}_{10}$	27_{10}

Worksheet 3: Binary Addition — Answer Key

Question	Answer
$101_2 + 11_2$	1000_2
$110_2 + 101_2$	1011_2
$111_2 + 111_2$	1110_2
$1010_2 + 101_2$	1111_2
$1101_2 + 1011_2$	11000_2

Worksheet 4: Binary Subtraction — Answer Key

Question	Answer
$101_2 - 10_2$	11_2
$110_2 - 101_2$	1_2
$1001_2 - 11_2$	110_2
$1100_2 - 101_2$	111_2
$10100_2 - 111_2$	1101_2

7. Creating Flowcharts

Example Flowchart 1: Decimal to Binary Conversion

```
START
↓
Enter decimal number (e.g., 2510)
↓
Divide by 2 repeatedly until quotient = 0
25÷2=12R1, 12÷2=6R0, 6÷2=3R0, 3÷2=1R1, 1÷2=0R1
↓
Record each remainder: 1, 0, 0, 1, 1
↓
Stop when quotient = 0
↓
Read remainders from BOTTOM to TOP
1, 1, 0, 0, 1
↓
This is your binary number: 110012
↓
END
```

Example Flowchart 2: Binary to Decimal Conversion

```
START
↓
Enter binary number: 11012
↓
Identify place values (powers of 2)
1, 2, 4, 8 (from right)
↓
Multiply each digit by its place value
1×8 + 1×4 + 0×2 + 1×1
↓
Add the results
8 + 4 + 0 + 1 = 13
↓
This is your decimal number: 1310
↓
END
```

Example Flowchart 3: Binary Addition

```
START
↓
Write numbers:  $101_2$  and  $11_2$  (column by column)
↓
Start from RIGHTMOST column
↓
 $1 + 1 = 2 \rightarrow$  write 0, carry = 1
↓
Next column:  $0 + 1 + \text{carry } 1 = 2 \rightarrow$  write 0, carry = 1
↓
Next column:  $1 + 0 + \text{carry } 1 = 2 \rightarrow$  write 0, carry = 1
↓
All columns done? YES
↓
Write final answer:  $1000_2$ 
↓
END
```

8. Creating Binary Patterns

Decimal	Binary	Pattern
0	0_2	Start
1	1_2	Add 1
2	10_2	Carry to next column ($1+1=10_2$)
3	11_2	Add 1 to 10_2
4	100_2	Carry to next column ($11_2+1_2=100_2$)
5	101_2	Add 1 to 100_2
6	110_2	Add 1 to 101_2
7	111_2	Add 1 to 110_2
8	1000_2	Carry to next column ($111_2+1_2=1000_2$)
9	1001_2	Add 1 to 1000_2
10	1010_2	Add 1 to 1001_2
11	1011_2	Add 1 to 1010_2
12	1100_2	Add 1 to 1011_2
13	1101_2	Add 1 to 1100_2
14	1110_2	Add 1 to 1101_2
15	1111_2	Add 1 to 1110_2
16	10000_2	Carry to next column

9. Creating Answer Keys

Example Answer Key for Decimal to Binary

Problem	Answer	Steps
$19_{10} \rightarrow \text{Binary}$	10011_2	$19 \div 2 = 9R1$, $9 \div 2 = 4R1$, $4 \div 2 = 2R0$, $2 \div 2 = 1R0$, $1 \div 2 = 0R1$
$28_{10} \rightarrow \text{Binary}$	11100_2	$28 \div 2 = 14R0$, $14 \div 2 = 7R0$, $7 \div 2 = 3R1$, $3 \div 2 = 1R1$, $1 \div 2 = 0R1$
$37_{10} \rightarrow \text{Binary}$	100101_2	$37 \div 2 = 18R1$, $18 \div 2 = 9R0$, $9 \div 2 = 4R1$, $4 \div 2 = 2R0$, $2 \div 2 = 1R0$, $1 \div 2 = 0R1$

Example Answer Key for Binary to Decimal

Problem	Answer	Steps
$11001_2 \rightarrow \text{Decimal}$	25_{10}	$1 \times 16 + 1 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1 = 16 + 8 + 0 + 0 + 1 = 25$
$10110_2 \rightarrow \text{Decimal}$	22_{10}	$1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1 = 16 + 0 + 4 + 2 + 0 = 22$
$11101_2 \rightarrow \text{Decimal}$	29_{10}	$1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 16 + 8 + 4 + 0 + 1 = 29$

10. Creating Problem-Solving Templates

Template 1: Fill in the Blanks (Decimal to Binary)

$$19_{10} = 10011_2$$

Step 1: $19 \div 2 = 9 \text{ R } 1$

Step 2: $9 \div 2 = 4 \text{ R } 1$

Step 3: $4 \div 2 = 2 \text{ R } 0$

Step 4: $2 \div 2 = 1 \text{ R } 0$

Step 5: $1 \div 2 = 0 \text{ R } 1$

Read bottom to top: 10011_2

Template 2: Fill in the Blanks (Binary to Decimal)

$$1101_2 = 13_{10}$$

Place values: 8, 4, 2, 1

$$1 \times 8 = 8$$

$$1 \times 4 = 4$$

$$0 \times 2 = 0$$

$$1 \times 1 = 1$$

$$\text{Sum: } 8 + 4 + 0 + 1 = 13_{10}$$

Template 3: Fill in the Blanks (Binary Addition)

$$\begin{array}{r} 101_2 \\ + 11_2 \\ \hline 1000_2 \end{array}$$

Right column: $1 + 1 = 2$ (write 0, carry 1)

Middle column: $0 + 1 + \text{carry } 1 = 2$ (write 0, carry 1)

Left column: $1 + 0 + \text{carry } 1 = 2$ (write 0, carry 1)

Final answer: 1000_2

11. Creating Self-Test Questions

Self-Test: Decimal to Binary — Answers

No.	Question	Answer
1	Convert 7_{10} to binary	111_2
2	Convert 14_{10} to binary	1110_2
3	Convert 21_{10} to binary	10101_2
4	Convert 36_{10} to binary	100100_2
5	Convert 45_{10} to binary	101101_2

Self-Test: Binary to Decimal — Answers

No.	Question	Answer
1	Convert 100_2 to decimal	4_{10}
2	Convert 1011_2 to decimal	11_{10}
3	Convert 11001_2 to decimal	25_{10}
4	Convert 101010_2 to decimal	42_{10}
5	Convert 111111_2 to decimal	63_{10}

Self-Test: Binary Addition & Subtraction — Answers

No.	Question	Answer
1	$101_2 + 11_2 =$	1000_2
2	$1101_2 + 1010_2 =$	10111_2
3	$1000_2 - 1_2 =$	111_2
4	$1100_2 - 11_2 =$	1001_2
5	$1111_2 + 1111_2 =$	11110_2

12. Creation Checklist — Completed

When creating binary numbers:

- Choose a decimal number
- Divide by 2 repeatedly
- Record all remainders
- Read from bottom to top
- Write with subscript 2
- Verify by converting back

When creating decimal numbers:

- Choose a binary number
- Identify place values (powers of 2)
- Multiply each digit by its place value
- Add all results
- Write with subscript 10
- Verify by converting back

When creating addition problems:

- Choose 2 or more binary numbers
- Add column by column
- Handle carries correctly
- Convert to decimal to verify
- Write the problem clearly

When creating subtraction problems:

- Choose two binary numbers
- Subtract column by column
- Handle borrowing correctly
- Convert to decimal to verify
- Write the problem clearly

13. Creation Templates Summary

1. DECIMAL → BINARY:

- Choose a decimal number (e.g., 19_{10})
- Divide by 2 repeatedly: $19 \div 2 = 9R1$, $9 \div 2 = 4R1$, $4 \div 2 = 2R0$, $2 \div 2 = 1R0$, $1 \div 2 = 0R1$
- Read remainders bottom to top: 10011_2

2. BINARY → DECIMAL:

- Choose a binary number (e.g., 1101_2)
- Multiply by powers of 2: $1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$
- Add the results: $8 + 4 + 0 + 1 = 13_{10}$

3. BINARY ADDITION:

- Choose two or more binary numbers: 101_2 and 11_2
- Add column by column: $101_2 + 11_2 = 1000_2$
- Carry when sum ≥ 2 : $1+1=2 \rightarrow$ write 0 carry 1

4. BINARY SUBTRACTION:

- Choose two binary numbers: 100_2 and 1_2
- Subtract column by column: $100_2 - 1_2 = 11_2$
- Borrow when 0-1: borrow from next column

5. WORKSHEET:

- 5+ questions on each topic
- Include answer key with steps

6. FLOWCHART:

- Visual steps for conversions
- Show decision points

14. Quick Creation Guide — Examples

Creating a Binary Number:

1. Pick a number: 23_{10}
2. Divide by 2 until 0: $23 \div 2 = 11R1$, $11 \div 2 = 5R1$, $5 \div 2 = 2R1$, $2 \div 2 = 1R0$, $1 \div 2 = 0R1$
3. Read remainders bottom to top: 10111_2
4. Check: $1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 1 \times 1 = 23 \checkmark$

Creating a Decimal Number:

1. Pick a binary number: 1101_2
2. Assign place values: 8, 4, 2, 1
3. Multiply: $1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 13$
4. Answer: 13_{10}

Creating an Addition Problem:

1. Pick 2 numbers: 111_2 and 11_2
2. Add column by column:
$$\begin{array}{r} 111_2 \\ + 11_2 \\ \hline 1010_2 \end{array}$$
3. Verify: $7+3=10=1010_2 \checkmark$

Creating a Subtraction Problem:

1. Pick 2 numbers: 100_2 and 1_2
2. Subtract column by column:

$$\begin{array}{r} 100_2 \\ - \quad 1_2 \\ \hline \end{array}$$

11_2
 3. Verify: $4-1=3=11_2$ ✓

Self-Test — Creation Questions — Answer Key

No.	Task	Answer
1	Create a decimal number and convert to binary	e.g., $25_{10} = 11001_2$
2	Create a binary number and convert to decimal	e.g., $10101_2 = 21_{10}$
3	Create a binary addition problem	e.g., $101_2 + 11_2 = 1000_2$
4	Create a binary subtraction problem	e.g., $1000_2 - 11_2 = 101_2$
5	Create a worksheet with 3 decimal→binary problems	e.g., $7_{10}=111_2$, $12_{10}=1100_2$, $19_{10}=10011_2$
6	Create a worksheet with 3 binary→decimal problems	e.g., $101_2=5_{10}$, $1101_2=13_{10}$, $10010_2=18_{10}$
7	Create a flowchart for binary addition	See example in Section 7
8	Create a binary counting pattern (0-16)	See example in Section 8
9	Create an answer key for 3 conversion problems	See example in Section 9
10	Create a multiple addition problem	e.g., $111_2 + 11_2 + 1_2 = 1011_2$

16. Creating Your Own Examples — Completed

My Binary Examples

Decimal	Division Steps	Binary
27_{10}	$27 \div 2 = 13R1$, $13 \div 2 = 6R1$, $6 \div 2 = 3R0$, $3 \div 2 = 1R1$, $1 \div 2 = 0R1$	11011_2
31_{10}	$31 \div 2 = 15R1$, $15 \div 2 = 7R1$, $7 \div 2 = 3R1$, $3 \div 2 = 1R1$, $1 \div 2 = 0R1$	11111_2
45_{10}	$45 \div 2 = 22R1$, $22 \div 2 = 11R0$, $11 \div 2 = 5R1$, $5 \div 2 = 2R1$, $2 \div 2 = 1R0$, $1 \div 2 = 0R1$	101101_2

My Binary to Decimal Examples

Binary	Place Values	Multiplication	Decimal
11001_2	16,8,4,2,1	$1 \times 16 + 1 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1$	25_{10}
10110_2	16,8,4,2,1	$1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1$	22_{10}
11100_2	16,8,4,2,1	$1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 0 \times 1$	28_{10}

My Binary Addition Examples

Problem	Column by Column	Answer
$110_2 + 101_2$	$0+1=1$, $1+0=1$, $1+1=2 \rightarrow 0$ carry1	1011_2
$111_2 + 111_2$	$1+1=2 \rightarrow 0$ carry1, $1+1+1=3 \rightarrow 1$ carry1, $1+1+1=3 \rightarrow 1$ carry1	1110_2

17. Create Your Own Binary Table — Completed

Decimal	Binary	Check
1	1_2	$1_2 = 1_{10}$ ✓
2	10_2	$10_2 = 2_{10}$ ✓
3	11_2	$11_2 = 3_{10}$ ✓
4	100_2	$100_2 = 4_{10}$ ✓
5	101_2	$101_2 = 5_{10}$ ✓
6	110_2	$110_2 = 6_{10}$ ✓
7	111_2	$111_2 = 7_{10}$ ✓
8	1000_2	$1000_2 = 8_{10}$ ✓
9	1001_2	$1001_2 = 9_{10}$ ✓
10	1010_2	$1010_2 = 10_{10}$ ✓
11	1011_2	$1011_2 = 11_{10}$ ✓
12	1100_2	$1100_2 = 12_{10}$ ✓
13	1101_2	$1101_2 = 13_{10}$ ✓
14	1110_2	$1110_2 = 14_{10}$ ✓
15	1111_2	$1111_2 = 15_{10}$ ✓
16	10000_2	$10000_2 = 16_{10}$ ✓

Binary Table — Quick Reference (1-32)

Decimal	Binary	Decimal	Binary
1	1 ₂	17	10001 ₂
2	10 ₂	18	10010 ₂
3	11 ₂	19	10011 ₂
4	100 ₂	20	10100 ₂
5	101 ₂	21	10101 ₂
6	110 ₂	22	10110 ₂
7	111 ₂	23	10111 ₂
8	1000 ₂	24	11000 ₂
9	1001 ₂	25	11001 ₂
10	1010 ₂	26	11010 ₂
11	1011 ₂	27	11011 ₂
12	1100 ₂	28	11100 ₂
13	1101 ₂	29	11101 ₂
14	1110 ₂	30	11110 ₂
15	1111 ₂	31	11111 ₂
16	10000 ₂	32	100000 ₂

Revision Tip: Always verify your binary answers by converting back to decimal. This is the fastest way to check if your work is correct!